W13: Sequence Learning Quiz

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* Indicates required question

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1. Which of the following best describes sequential data? 1 point

- A) Data with no dependencies between data points
- B) Data where the order of elements is crucial to its meaning
- C) Data that is always formatted in a relational database
- D) Data consisting exclusively of independent numerical values

2. Which of these is a classic example of time series data? 1 point

- A) A cross-sectional survey of a population
- B) A folder of independent, unordered images
- C) Daily stock market closing prices over a year
- D) The final score of a single basketball game

3. What is the primary advantage of Recurrent Neural Networks (RNNs) over standard Feedforward Neural Networks? 1 point

- A) They process spatial data with filters
- B) They do not require backpropagation to learn
- C) They maintain a hidden state that captures information from previous time steps
- D) They train significantly faster on large datasets

4. What is the "vanishing gradient" problem commonly faced by basic RNNs? 1 point

- A) Gradients grow exponentially, causing the model to diverge
- B) The loss function becomes zero during the first epoch
- C) Gradients shrink exponentially during backpropagation through time, making it hard to learn long-range dependencies
- D) The model forgets to update its bias weights

5. Exploding gradients in RNNs are most commonly mitigated by which technique? 1 point

- A) Gradient clipping
- B) Removing all activation functions
- C) Increasing the learning rate
- D) Adding more recurrent layers

6. In a Long Short-Term Memory (LSTM) network, which gate decides what information to discard from the cell state? 1 point

- A) Input gate
- B) Output gate
- C) Update gate
- D) Forget gate

7. What is a key structural difference between a Gated Recurrent Unit (GRU) and an LSTM? 1 point

- A) GRUs have more gates than LSTMs
- B) GRUs combine the forget and input gates into a single update gate
- C) GRUs do not use a hidden state
- D) GRUs rely exclusively on self-attention

8. In an LSTM, what is the primary role of the "cell state"? 1 point

- A) To output the final prediction directly to the user
- B) To act as a conveyor belt, carrying relevant information across sequence steps with minimal modification
- C) To reset the neural network's weights after each batch
- D) To replace the hidden state entirely

9. Which of the following gates is NOT present in a standard LSTM architecture? 1 point

- A) Forget gate
- B) Input gate
- C) Update gate
- D) Output gate

10. A standard Sequence-to-Sequence (Seq2Seq) model consists of which two primary components? 1 point

- A) Generator and Discriminator
- B) Encoder and Decoder
- C) Actor and Critic
- D) Convolutional and Pooling layers

11. In a basic Seq2Seq model used for Machine Translation, what does the Encoder output and pass to the Decoder? 1 point

- A) A fixed-length context vector (thought vector)
- B) The raw input text
- C) A probability distribution of the entire vocabulary
- D) A matrix of independent state vectors

12. What is a major bottleneck of the basic Seq2Seq architecture without attention? 1 point

- A) It cannot handle fixed-length inputs
- B) The fixed-length context vector struggles to compress all necessary information from long input sequences
- C) The decoder generates tokens too quickly to process
- D) It only works with binary classification tasks

13. Which of the following is a classic example of a sequence-to-sequence learning task? 1 point

- A) Image classification
- B) Text summarization
- C) K-means clustering
- D) Principal Component Analysis

14. How does Greedy Search select the next word during decoding? 1 point

- A) It explores multiple sequences and picks the most likely complete sentence
- B) It randomly samples a word based on a temperature distribution
- C) It picks the single word with the highest probability at the current step
- D) It selects the word that has the lowest global loss

15. What is the primary limitation of Greedy Search decoding? 1 point

- A) It requires a massive memory footprint
- B) It is computationally too expensive for real-time translation
- C) It may miss the optimal overall sequence by only focusing on immediate best choices
- D) It cannot be used with RNNs

16. How does Beam Search improve upon Greedy Search? 1 point

- A) It calculates the probability of every single sequence possible in the vocabulary
- B) It generates the entire sequence at once instead of step-by-step
- C) It keeps track of the top 'k' most probable sequences at each step
- D) It eliminates the need for a Decoder network

17. If a Beam Search algorithm uses a beam width of 1 (k=1), how does it behave? 1 point

- A) It fails to generate any output
- B) It generates completely random sequences
- C) It behaves exactly like Greedy Search
- D) It selects only the most frequent word in the training data

18. What core problem of basic Seq2Seq models does the Attention mechanism solve? 1 point

- A) It eliminates the need for mathematical gradients
- B) It allows the decoder to dynamically focus on different parts of the input sequence for each output step
- C) It further compresses the sequence into a smaller fixed-length vector
- D) It replaces all activation functions with linear transformations

19. The seminal paper "Attention Is All You Need" introduced which highly influential deep learning architecture? 1 point

- A) The Long Short-Term Memory (LSTM) network
- B) The Convolutional Neural Network (CNN)
- C) The Transformer
- D) The Generative Adversarial Network (GAN)

20. Unlike traditional RNNs, how do Transformers process sequence data? 1 point

- A) Strictly left-to-right, one token at a time
- B) Strictly right-to-left, one token at a time
- C) In parallel, processing all tokens simultaneously using self-attention mechanisms
- D) By compressing all tokens into a single recurrent cell state

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